b. External growler

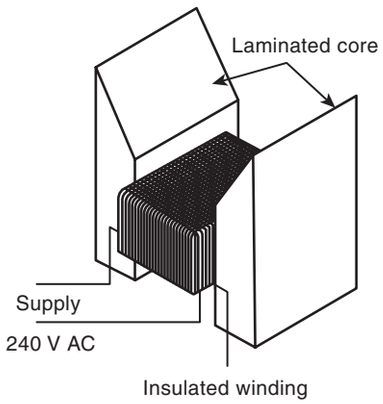


Fig. 10.6 External growler

External growler consists of a winding over 'H' shaped laminated iron core. One side of its leg is cut out on the top in slantingly to place the rotor or armature winding.

c. Working principle

When we give AC, 1ϕ, 230v, supply to the growler winding, it works in transformer principle (Mutual induction). Growler winding acts like a primary of the transformer and the winding to be tested acts like secondary. Due to this, magnetic field voltage is induced.

For testing stator and rotor windings of AC motors, hacksaw blade is used and the fault can be identified by its vibrations.

For testing the armature windings of DC machines AC mili voltmeter is connected with the commutator segment and the fault can be identified by readings.

d. Testing of armature winding by growler

i. Open circuit test

The armature to be tested is placed on the growler as shown in the figure 10.7. AC supply is given to the growler. Now voltage is induced in the armature windings according to the transformer action and it can be measured by AC milli voltmeter. The two leads of the AC milli voltmeter is connected with the adjacent bars as shown in the figure 10.7.

The pointer of the milli voltmeter must deflect to any value when we made test on all commutator segments. If the pointer does not deflect, the coil connected with the particular commutator segment is subjected to open circuit fault. The fault may be inside the coil or in the soldering part of the commutator segment.

This test can be made without the meter. For example, by shorting the top two bars with a piece of wire instead of meter. If there is a spark, it indicates the armature coil is in good condition. Absence of spark indicates that the coil is open.

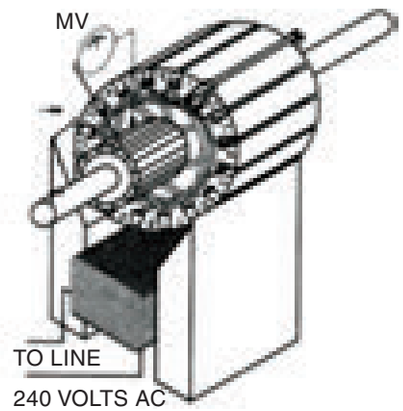


Fig. 10.7 Open circuit test

ii. Short circuit test

The armature to be tested is placed on the growler as shown in the figure and AC supply is given to the growler. Hold a hacksaw blade over the top slots of the armature as shown in the figure 10.8. If the coil in the slot is shorted, the blade will vibrate rapidly and create a growling noise. If the blade remains stationary, it is an indication that no short exist in the coil under test.

BEE_ENG_Chapter_10.indd 163 12/27/2019 10:39:27 AM